

Java Engineer / Salesforce Developer



Assignment

We want to make your interview process mutually beneficial. This is an opportunity for us to assess your ability to solve business gaps using Salesforce Solutions, and for you to gain a better understanding of what this role ultimately means.

We leave the format of your documentation & presentation up to you. You can choose to prepare slides, go for a casual conversation, prepare a demo or decide on other creative approaches. At Adyen, we value content over form so do what makes you feel more comfortable. However, please submit your content to us at least 2 days before the case interview to salesforcedev@adyen.com.

Feel free to make assumptions in your presentation if you need to - just make sure to list them at the beginning so the interviewers are aware.

Below you find an guideline of the interview structure, this is not set in stone:

- 5 minutes introduction
- 15 minutes case presentation
- 60 minutes Q&A and Technical questions
- 10 minutes for questions from the candidate

Assignment 1: The Adyen Animal Shelter

As Adyen, we would like to start a new sub company that can take care of animals when they are abandoned, called *Adyen Animal Shelter*.

As a first step, we would like to start with helping abandoned dogs currently sheltered at an associated shelter facility; finding them a new home through an adoption process. Adyen decided that Salesforce is the right tool for driving the *Adyen Animal Shelter* project to success.

Your assignment

As the company developer, your assignment is to design and implement a solution for this initial stage of the project:

1. Help the organization to automatically add new Dogs to the system and use an external source for fetching their pictures.
2. Design and implement a Data Model for supporting an "Adoption Lifecycle" that follows the process from *created* to *adopted*.
3. Create a component that lists and displays all dogs available for adoption.
4. Prepare some documentation that includes:

- detailed instructions for your solution.
- explanation of the technical choices you took.
- Any point you think is relevant.

Requirements

- When an agent creates a new “record” in Salesforce, they need to specify the abandoned animal and its information.
- Upon record creation:
 - Salesforce should notify an external system (details below) that will provide the dog image for the record, using the “fetch random image by breed” functionality.
 - Salesforce forwards the adoption request to the associated shelter facility Database.

Future Improvements

- *Adyen Animal Shelter* has plans to grow in the future, starting new associations with multiple regional shelter facilities around the country.
- Provide a scalable and future-proof architectural design that will allow the Salesforce instance to scale up: the newly created adoption requests will be forwarded to one among the many different Animal Shelter regional Databases (it is not required to actually implement the solution).

Assignment 2: Architectural Challenge: India Expansion & Data Residency

As a next step in its global expansion, the Adyen Animal Shelter plans to open a new, independent shelter facility in India. This new shelter will manage its own set of animals and adoptions.

Due to local data privacy regulations, all personally identifiable information (PII) related to the animals and the individuals adopting them must remain stored on infrastructure within India. This data is not permitted to be stored in the global Salesforce instance.

Your Assignment:

Your task is to design a scalable and compliant architectural solution that allows your global Salesforce users to manage the India shelter's operations while adhering to this strict data residency requirement.

Please prepare the following as part of your presentation:

1. **Architecture Diagram:** Create a high-level diagram illustrating your proposed solution. The diagram should clearly show:
 - a. The global Salesforce instance.
 - b. The India-based data store for sensitive information.
 - c. The middleware/service layer that connects them.

- d. The flow of data for key operations (e.g., creating a new animal record, viewing an animal's details).
2. **Technical Explanation:** Be prepared to discuss the technical choices behind your design, including:
 - a. **Middleware:** What technology stack would you recommend for the service layer that acts as a gateway between Salesforce and the India database? Explain your reasoning, keeping in mind the skills required for this role (e.g., Java, Spring Boot, REST/gRPC APIs).
 - b. **Data Flow:** How would a user in the global Salesforce org view the details of a dog from the India shelter? Explain how your solution prevents the animal's sensitive data from being stored outside of India while still providing a seamless user experience.
 - c. **Scalability:** How does your design support future expansion to other regions with similar data residency requirements?

Technical Details

- Do this challenge in any Salesforce developer sandbox.
- You can use a second Salesforce for "mocking" the associated shelter facility Database.
- Follow this API documentation for the Dogs details: [Dog CEO Api documentation](#)
 - The JSON Object structure is simple:

```
{"message": "IMG_URL", "status": "STATUS"}
```
- Scalability and future improvements: keep in mind the option to scale up the *Adyen Animal Shelter* and make decisions depending on that.

Guidelines

- You have complete freedom to do what you think is necessary for your solution.
- Feel free to make assumptions on what has not been listed in the requirements, creativity is appreciated!
- Use any resources you need, but refrain from the help of others, it is important that you do the assessment on your own.
- Create an admin user for salesforcedev@adyen.com **at least 2 working days** before the skills interview.

If you have any questions about this case study, please do not hesitate to reach out. Good luck and we can't wait to see what you create!